

Section : Physics

- Q.1** An electric dipole is placed in a non-uniform electric field, such that the dipole moment is parallel to the electric field. The dipole will experience
1. a torque only.
 2. both torque and force .
 3. neither force nor torque
 4. a force only

Options

1. 1
2. 2
3. 3
4. 4

- Q.2** A solenoid with an iron core and a bulb in series are connected to a d.c. source. When the iron core is removed from the solenoid, the brightness of the bulb will
1. increase
 2. decrease
 3. remain unaffected
 4. cease to zero

Options

1. 1
2. 2
3. 3
4. 4

- Q.3** Which of the following statements are correct?
- (A) Kirchhoff's loop rule represents energy conservation.
(B) Resistivity is a property of the material and depends on temperature.
(C) The electrical conductivity of the material does not depend on the material of the conductor and its temperature.
(D) The emf of a battery equals the potential difference between its terminals when the terminals are connected externally.
- Choose the correct answer from the options given below:
1. (A) and (D) only
 2. (A), (B) and (C) only
 3. (A) and (B) only
 4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.4** A circular loop of area $A = 0.01 \text{ m}^2$ carrying a current of $I = 10 \text{ A}$, is held in a magnetic field of $B = 0.1 \text{ T}$ parallel to the plane of the loop. The torque acting on the loop is-
1. 0.001 N m
 2. 0.01 N m
 3. 0.1 N m
 4. zero

Options

1. 1
2. 2
3. 3
4. 4

- Q.5** In a Hydrogen atom there is transition of electron from one orbit to another. Choose the correct option given below in which the emitted photon will have the highest frequency?
1. $n = 3$ to $n = 2$
 2. $n = 6$ to $n = 2$
 3. $n = 2$ to $n = 1$
 4. $n = 4$ to $n = 2$

Options

1. 1
2. 2
3. 3
4. 4

- Q.6** A gaussian surface encloses three point charges $q_1 = -14 \text{ nC}$, $q_2 = 78.85 \text{ nC}$ and $q_3 = -56 \text{ nC}$. The electric flux for the gaussian surface due to these charges is
1. $1000 \text{ N m}^2 \text{ C}^{-1}$
 2. $7885 \text{ N m}^2 \text{ C}^{-1}$

3. $885 \text{ N m}^2 \text{ C}^{-1}$

4. $1400 \text{ N m}^2 \text{ C}^{-1}$

Options

1. 1
2. 2
3. 3
4. 4

Q.7 A cylindrical, straight long wire having radius ' a ' is carrying current ' I ' uniformly distributed inside the wire. As per Ampere's circuital law, the magnetic field inside the wire at any distance r , ($r < a$) is given by:

1. $B = \mu_0 I / 2\pi r$
2. $B = \mu_0 I a / 2\pi r$
3. $B = \mu_0 I r / 2\pi a^2$
4. $B = \mu_0 I a / 2\pi r^2$

Options

1. 1
2. 2
3. 3
4. 4

Q.8 Two metals, X and Y have work functions, 4 eV and 10 eV, respectively. The ratio of threshold wavelengths of metal X to metal Y is

1. 5:2
2. 1:1
3. 2:5
4. 4:25

Options

1. 1
2. 2
3. 3
4. 4

Q.9 The ratio of nuclear densities of two nuclei having mass numbers A and $8A$, respectively is

1. 1:2
2. 1:1
3. 1:8
4. 2:1

Options

1. 1
2. 2
3. 3
4. 4

Q.10 An electron is accelerated through a potential difference of $V = 36 \text{ V}$. The de-Broglie wavelength associated with it will be

1. $2.05 \times 10^{-10} \text{ m}$
2. $2.26 \times 10^{-9} \text{ m}$
3. $1.26 \times 10^{-10} \text{ m}$
4. $2 \times 10^{-7} \text{ m}$

Options

1. 1
2. 2
3. 3
4. 4

Q.11 Choose the incorrect statements from the options given below:

- (A) In making a full wave rectifier, the two diodes are connected, with one in forward bias and the second in reverse bias.
- (B) In a full wave rectifier, the output frequency is twice the input frequency.
- (C) In a full wave rectifier, the current through the load resistance flows in one direction for half cycle and in the reverse direction for the other half cycle.
- (D) In a full wave rectifier having two diodes and a resistance, the output voltage is uniform.
- Choose the correct answer from the options given below:
1. (A) and (B) only
 2. (A) and (C) only
 3. (A) and (D) only
 4. (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.12 An electron of mass 9.1×10^{-31} kg carrying charge 1.6×10^{-19} C, is moving horizontally with a speed of 4×10^6 m/s in a uniform magnetic field of magnitude 8×10^{-4} T acting vertically upward. The radius of its circular path is,

1. 28.4 cm
2. 2.84 cm
3. 24 cm
4. 32 cm

Options

1. 1
2. 2
3. 3
4. 4

Q.13 A positively charged particle moving along $+X$ axis, enters a region of uniform magnetic field directed vertically upward in $+Z$ direction. The particle will _-_-_

1. move in a circular path with an increased speed
2. move in a circular path with a uniform speed
3. move in a circular path with a decreased speed
4. get deflected in a vertically upward direction

Options

1. 1
2. 2
3. 3
4. 4

Q.14 In order to fabricate n-type semiconductor, the impurity elements used are

1. acceptor atoms like Aluminum, Indium.
2. donor atoms like Phosphorous, Arsenic.
3. carbon as it has four valence electrons.
4. semiconductor atoms like Silicon and Germanium.

Options

1. 1
2. 2
3. 3
4. 4

Q.15 In a coil of self inductance 10 H, the rate of change of current is 4 A s^{-1} . The electromotive force (emf) induced in the coil will be

1. -2.5 V
2. 2.5 V
3. 40 V
4. -40 V

Options

1. 1
2. 2
3. 3
4. 4

Q.16 Choose the incorrect statements from the options given below:

- (A) If a monochromatic source is replaced by a source of white light in Young's experiment, then interference fringes will be coloured,
(B) If the screen is moved away from the plane of slits, then the angular width of the fringe will not remain the same,
(C) If separation between slits in Young's experiment is halved, the fringe width becomes double,
(D) The fringe width in Young's experiment is $\lambda d/D$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (D) only
3. (A) and (C) only
4. (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.17** A parallel plate air capacitor has a capacitance C . When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be
- 1.200%
 - 2.400%
 - 3.66.6%
 - 4.33.3%

Options

- 1
- 2
- 3
- 4

- Q.18** There exists a horizontal magnetic field in space of value $B = 9 \times 10^{-4}$ T. A magnetic bar of magnetic moment $M = 9 \times 10^3$ J/T is placed at an angle $\theta_i = 60^\circ$ to the field and is free to rotate. Finally, it comes to a position parallel to the field. Work done by magnetic field on the magnetic bar will be

1. 1.4×10^{-2} J
2. -40.5 J
3. -4.05 J
4. 4.05 J

Options

- 1
- 2
- 3
- 4

- Q.19** The property of a medium which determines the electric force between two charges situated in the medium is
1. Magnetic susceptibility
 2. Permittivity
 3. Conductivity and permeability
 4. Permeability only

Options

- 1
- 2
- 3
- 4

- Q.20** Rohit has been given two converging lenses of focal lengths 1.25 cm and 5 cm, respectively to design a compound microscope. If it is desired to have a magnification of 30, what would be the separation between the objective and the eyepiece when the final image is formed at the least distance of distinct vision?

1. 5.0 cm
2. 7.50 cm
3. 6.25 cm
4. 4.30 cm

Options

- 1
- 2
- 3
- 4

- Q.21** Factors, on which the capacitance of a parallel plate capacitor depends, are
- (A) Charge on the plates
 - (B) Area of plates
 - (C) Permittivity of the medium between the plates
 - (D) Distance between the plates

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

- 1
- 2
- 3
- 4

- Q.22** Which of the following statements are correct?
- (A) In a p-n junction diode, accumulation of electric charge of the opposite type on two sides of the junction produces an electric field and a potential barrier,
 - (B) the potential barrier opposes the movement of the majority charge carriers across the junction.
 - (C) In the reverse biased p-n junction, the width of the depletion layer decreases.
 - (D) In forward bias of a p-n junction, the potential barrier increases.
- Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A) and (C) only
3. (A) and (B) only
4. (C) and (B) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.23** Two identical point charges placed at a distance 'r' apart exert a force F on each other. If each charge is halved and the distance between them is doubled, then the new force acting on each charge would be
1. 4F
 2. F/4
 3. F/16
 4. 8F

Options

1. 1
2. 2
3. 3
4. 4

- Q.24** A pair of adjacent coils has a mutual inductance of 2.0 H . If the current in one coil changes from 0 to 30 A in 0.5 s, the change of the magnetic flux linked with the other coil would be
1. 30 Wb
 2. 120 Wb
 3. 50 Wb
 4. 60 Wb

Options

1. 1
2. 2
3. 3
4. 4

- Q.25** Choose the incorrect statements from the options given below:
- (A) S I unit of inductive reactance and capacitive reactance is Ohm
 - (B) Inductive reactance acting like resistance, stops the dc and ac both
 - (C) Inductive reactance and capacitive reactance both are proportional to frequency of ac supply,
 - (D) There is no power consumption in pure capacitive reactance.
- Choose the correct answer from the options given below:
1. (A) and (B) only
 2. (A) and (C) only
 3. (B) and (C) only
 4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.26** The electric field and magnetic field of a plane electromagnetic wave propagating in vacuum are expressed as
- $$E_x = E_0 \sin(kz - \omega t)$$
- $$B_y = B_0 \sin(kz - \omega t)$$
- Which of the following statements are correct?
- (A). The wave is propagating along the z- direction.
 - (B). The electric field E_x and magnetic field B_y are perpendicular to each other and perpendicular to the z- direction.
 - (C) The electric and magnetic fields E_x and B_y are perpendicular to each other, and parallel to the direction of propagation.
 - (D) k is related to the wave length λ of the wave by the usual equation $k = 2\pi/\lambda$
- Choose the correct answer from the options given below:
1. (A), (B) and (D) only
 2. (A) and (B) only
 3. (A) and (C) only
 4. (A), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.27** A square shaped coil of 10 cm side, having 100 turns, is placed perpendicular to a magnetic field which is increasing at the rate of 1.0 T/s. The induced emf in the coil would be
1. 0.75 V
 2. 1.0 V
 3. 0.5 V
 4. 0.1 V

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The mobility of charge carriers increases with

1. increase in the mass of charge carriers

2. increase in the electric field

3. decrease in the electric field

4. increase in the average collision time

Options 1. 1

2. 2

3. 3

4. 4

Q.29 A biconvex lens has radii of curvature of 10 cm each. If the refractive index of the material of the lens is 1.5, what will be its focal length in air ?

1. 5 cm

2. 10 cm

3. 20 cm

4. ∞

Options 1. 1

2. 2

3. 3

4. 4

Q.30 The electric potential at any point in space is given by $V = 4x^2$ volt. The electric field at the point (1 m, 0, 2 m) in SI unit is

1. 16, along the negative X - axis

2. 17.8, along the positive X - axis

3. 8, along the negative X - axis

4. 8, along the positive X - axis

Options 1. 1

2. 2

3. 3

4. 4

Q.31 When a convex lens of glass is immersed in water, its power will

1. increase

2. decrease

3. remain the same

4. become zero

Options 1. 1

2. 2

3. 3

4. 4

Q.32 If $\vec{M}$ is magnetic moment and $\vec{B}$ is magnetic field, then the torque ($\vec{\tau}$) is given by

1. $\vec{M} \cdot \vec{B}$

2. $\vec{M} \times \vec{B}$

3. $\vec{B} \times \vec{M}$

4. $\frac{|\vec{M}|}{|\vec{B}|}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 In Young's double slit experiment, the width of one of the two slits is four times the width of the other slit. The ratio of the maximum to the minimum intensity in the interference pattern will be

1. 4:1

2. 3:1

3. 9:1

4. 16:1

Options 1. 1

2. 2
3. 3
4. 4

Q.34 An electric motor is driven by d.c. supply of emf V_0 and internal resistance r . The current when output power is maximum will be

1. $4V_0/r$
2. V_0/r
3. $2V_0/r$
4. $V_0/2r$

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Which of the following statements are correct?

(A) In an ac circuit containing only resistance, power consumption is $I^2 R$.

(B) In an LCR circuit, there is resonance at zero phase difference between X_C and X_L and the power consumption is $I(\omega X_C + \omega X_L)^{1/2}$.

(C) In an LCR circuit, at resonance, impedance is minimum.

(D) In an LCR circuit, there is resonance at zero phase difference between X_C and X_L and the power consumption is zero.

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A) and (B) only
3. (A) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.36 A heating element is marked $V = 210 \text{ V}$, $P = 630 \text{ W}$. The resistance of the element (R) would be

1. 50Ω
2. 60Ω
3. 40Ω
4. 70Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.37 The extraordinary sparkling of an appropriately cut diamond is due to

1. dispersion
2. polarisation
3. low refractive index
4. Total internal reflection

Options 1. 1

2. 2
3. 3
4. 4

Q.38 The energy of a photon of wavelength (λ) is given by the expression:

(Symbols have their usual meaning: h = Planck's constant, c = speed of light)

1. $h\lambda c^{-1}$
2. $h\lambda^{-1}c^{-1}$
3. $h\lambda^{-1}c$
4. $h^{-1}\lambda c$

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Choose the **correct** statements from the following:

- (A) Electromagnetic waves do not follow the principle of superposition.
- (B) The variations in the electric field and in the magnetic field of electromagnetic waves, if any, never happen simultaneously.
- (C) Electromagnetic waves are produced by accelerated charges.
- (D) The energy of electromagnetic waves is divided equally into electric and magnetic field vectors.

Choose the **correct** answer from the options given below:

- 1. (A) and (B) only
- 2. (B) and (D) only
- 3. (A) and (C) only
- 4. (C) and (D) only

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.40 A long straight wire carries a current of 12 A. Another straight wire of length 5 cm placed parallel to it at a distance of 5 cm, carries a current of 2 A. The magnetic force on the smaller wire is

- 1. 3.4×10^{-6} N
- 2. 4.8×10^{-6} N
- 3. 2.4×10^{-6} N
- 4. 9.6×10^{-7} N

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.41 Which of the following statements related to Wheatstone Bridge are correct?

- (A) The bridge is most sensitive when all the four resistances are of the same order
- (B) The internal resistance of the cell and the resistance of the galvanometer affect the null point
- (C) It does not involve measurement of current and potential difference
- (D) Resistances of ammeters and voltmeters do not affect the measurements

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (A), (C) and (D) only

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.42 A light of wavelength of $\lambda = 6 \times 10^{-7}$ m illuminates a single slit of width $a = 1.2$ mm. The screen is at a distance of $D = 2.4$ m away from the slit. The width of central maximum (β_0) is

- 1. 2.4 mm
- 2. 1.2 mm
- 3. 3.5 mm
- 4. 4.8 mm

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.43 Choose the **incorrect** statements from the following:

- (A) The energy density of an electric field in electromagnetic waves is $(\epsilon_0 E^2/2)$.
- (B) The total energy of electromagnetic waves falling on a surface is $E = 6.0 \times 10^5$ J and the momentum delivered to the surface, if absorbed fully, is $p = 2 \times 10^3$ kg m/s.
- (C) The speed of the electromagnetic waves is E_0/B_0 .
- (D) The energy density of magnetic field in electromagnetic Waves is (B^2/μ) .

Choose the correct answer from the options given below:

- 1. (A) and (B) only
- 2. (A) and (C) only
- 3. (B) and (D) only
- 4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.44 The angular momentum of an electron in the third orbit of a hydrogen atom is
(Given $h = 6.6 \times 10^{-34} \text{ J s}$)

1. $3.15 \times 10^{-34} \text{ J s}$

2. $5.13 \times 10^{-34} \text{ J s}$

3. $3.15 \times 10^{-36} \text{ J s}$

4. $5.15 \times 10^{-36} \text{ J s}$

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Which of the following statements are true for the minimum deviation position of a prism?

(A). The angle of the prism is twice the angle of refraction.

(B). The refracted ray is parallel to the base of prism.

(C). The angle of incidence is equal to the angle of emergence.

(D). The angle of deviation is half the angle of a prism.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A charge of $Q = 8.0 \times 10^{-9} \text{ C}$ is uniformly distributed on the surface of a spherical shell of radius $R = 10 \text{ cm}$. The electric field at a distance of 10 cm from the outer surface of the shell is

1. $7.2 \times 10^3 \text{ N/C}$

2. $7.2 \times 10^3 \text{ N/C}$

3. $18 \times 10^2 \text{ N/C}$

4. $1.8 \times 10^3 \text{ N/C}$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Three identical resistors are connected first in series to an electrical voltage V . Let I be the total current flowing in the circuit. These resistors are disconnected and again connected in parallel with the same source. Which of the following statements are true?

(A) The potential difference across each resistor is $V/3$ in series connection

(B) The potential difference across all three resistors in parallel connection is $3V$

(C) The current flowing in each resistor in series connection is the same as I

(D) The current flowing through each resistor in parallel combination is equal to $3I$

Choose the correct answer from the options given below:

1. (A) and (B) only

2. (A) and (C) only

3. (A), (C) and (D) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Eight cells each having emf 2 V and internal resistance $r = 0.02 \Omega$ are connected in series with an external resistance 9Ω . The current flowing through the external resistance (I) is

1. 1.60 A

2. 0.99 A

3. 1.75 A

4. 1.88 A

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.49 The ratio of maximum frequency and minimum frequency of light emitted in Balmer series of hydrogen spectrum is

- 1. 11:9
- 2. 9:5
- 3. 11:7
- 4. 16:7

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.50 In an a.c. circuit, a resistor, a capacitor and an inductor are connected in series, in which $X_C = X_L = R$. The impedance of this circuit will be

- 1. $R\sqrt{5}$
- 2. $R\sqrt{2}$
- 3. R
- 4. zero

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4